



BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT
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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE
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AI FUSION CHALLENGE- HACKATHON

Problem Statement -2

Problem Statement : An Intelligent Q&A Assistant

Title: “AI Vaidya: Building an Intelligent Q&A Assistant for Ayurveda Knowledge”

Problem Overview:

Ayurveda encompasses a vast body of traditional medical wisdom, spanning books, scriptures, and research documents.

However, this knowledge is often **unstructured, difficult to access, and not searchable in modern ways.**

Your challenge is to design an **AI-powered system** that can:

- Ingest text data (e.g., Ayurveda books, research papers, articles)
- Understand the content, and
- Answer user queries in natural language (English) based on that knowledge — **without using the internet** for answers.

Essentially, create a **domain-specific retrieval-based Q&A system** that acts as a digital Ayurvedic assistant — “AI Vaidya.”

Objective:

Develop a prototype where:

1. The system can **read and process Ayurvedic text** (books or datasets



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provided).

2. Users can **ask health or concept-based questions**, e.g.
 - “What are the doshas in Ayurveda?”
 - “How does turmeric help in wound healing?”
3. The system returns **context-aware answers** only from the uploaded text or dataset.

Expected Features:

- user interface (web) to ask questions.
- Backend that stores and retrieves relevant text chunks.
- Integration with **free APIs** for embeddings or NLP.
- Use of **vector search** or **semantic retrieval** to find relevant answers.

Technical Hints:

Data Input

Use an open-access Ayurveda text from:

- National Institute of Ayurveda open sources
- AYUSH research papers
- Or plain text/PDFs (can be downloaded from the internet)

Functional Flow:

1. Upload or load an Ayurveda book (text or PDF- min 400 pages)
2. System preprocesses and embeds text
3. User types a question → system retrieves top relevant chunks
4. The model generates the answer from those retrieved sections
5. Display answer + reference text snippet



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Example Questions:

- “What are the three Gunas in Ayurveda?”
- “How is digestion described in Ayurveda?”
- “Which herbs are useful for cough and cold according to classical texts?”

Hackathon Submission Guidelines

AI Vaidya – Ayurveda Intelligence Challenge

What to Submit

1. Working Application

- Submit a functional prototype that fulfils the hackathon problem statement requirements — i.e., an AI system capable of understanding Ayurvedic text and answering user queries.
- The application may be built as a web app, a desktop app, or a mobile prototype.
- It must demonstrate the following:
 - Ingestion of Ayurveda text/books/datasets.
 - Intelligent question answering based on uploaded data.
 - Integration of free/open APIs or open-source NLP models.

2. Text Description / Documentation

Each team must include a written description (maximum 2–3 pages or in the README) that clearly explains:



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- **Project Title and Problem Statement**

Describe the problem your solution addresses within the Ayurveda domain.

- **Solution Overview**

Summarise your application, its features, and how it solves the problem.

- **Technical Architecture**

Outline how your system works — including data processing, model usage, and user flow.

- **APIs / Libraries Used**

Mention all free APIs and open-source models used.

- **Future Scope**

Optional: suggest how the system can be scaled or enhanced.

3. Demonstration Video

Each team must provide a short demo video (maximum 3 minutes) that includes:

- A walkthrough of the working application.
- Footage showing key functionality (uploading text, asking questions, getting responses).
- Clear narration or captions explaining what's being shown.
- Video must be publicly viewable on YouTube or Vimeo.

Note:

- The video must not contain any copyrighted or infringing material.
- Include the public video link in your submission form.



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4. GitHub Repository

Submit a public GitHub repository containing your project source code. The repository must include:

- Complete source code of the working application.
- A README.md file with:
 - Project overview and setup instructions.
 - API/model usage documentation.
 - Steps to Run or Test the Application.
 - Any login credentials (if a private demo site is used).
- An open-source license (MIT, Apache 2.0, etc.).

5. Access for Testing

- Judges must be able to access and test your application.
- You can provide:
 - A public live demo URL, or
 - A hosted app link (Streamlit, Hugging Face Spaces, Render, etc.).
- If login access is required, include test credentials in the README.
- The application must remain publicly accessible until results are announced.

6. Language & Format

- All submissions (documentation, video, and app interface) must be in English.
- Code comments and variable names should be clear and readable.



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Submission Summary Checklist

Item	Description
Working Application	Functional prototype fulfilling hackathon requirements
Text Description	Project write-up + APIs used + functionality explained
Demo Video	< 3 mins, uploaded on YouTube/Vimeo (public link)
GitHub Repo	Public repo with open-source license & setup guide
Access Link	Website/demo link or deployment credentials

Evaluation Criteria

Category	Description	Weight
Innovation	Novelty of the approach / creativity	25%
Technical Implementation	Use of NLP, embeddings, and retrieval	25%
Accuracy	Answers based on source text (not hallucinated)	25%
UI/UX & Presentation	User-friendly design and explanation	25%

Expected Deliverable

A working prototype (web or desktop app) that demonstrates:

- Data ingestion (PDF or text)
- Retrieval + answer generation
- Interactive Q&A interface